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Sketch-Based Pattern Spotting in Historical Documents via Conditioned One-Stage Detection

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Abstract

Pattern spotting in historical documents has traditionally relied on an example image of the target pattern as the query, an assumption that fails whenever researchers and archivists have only a mental or sketched concept of what they seek. We address this limitation by proposing a sketch-conditioned, one-stage detector that localizes recurring graphical patterns in medieval document pages using a single hand-drawn sketch as the sole query. The model couples a frozen Vision Transformer backbone with a Feature Pyramid Network and an anchor-free FCOS detection head conditioned on the sketch through two complementary mechanisms: a global Feature-wise Linear Modulation (FiLM) layer that injects a coarse class-level prior into every pyramid level, and a bidirectional co-attention module that enables fine-grained spatial interaction between the sketch tokens and the document features. We compare two backbone choices iDoc, a ViT encoder domain-adapted to medieval manuscripts via LoRA fine-tuning on the HORAE corpus, and DINOv3, a general-purpose encoder on a dataset of 544 historical document pages annotated with 1,349 bounding boxes spanning 22 classes of recurring graphical patterns. DINOv3-FCOS achieves a mAP of 0.701 at IoU = 0.50, outperforming iDoc-FCOS (0.594) by +10.7 points, with consistent gains in precision (0.821 vs. 0.743), recall (0.865 vs. 0.757), and F1 (0.842 vs. 0.750). The advantage of DINOv3 is most pronounced for rare and visually complex classes, suggesting that representational breadth from large-scale general pre-training transfers more effectively to the trainable detection head than domain-specific adaptation, particularly in low-resource regimes. These results demonstrate the viability of sketch-based querying for historical document pattern spotting and highlight the critical role of backbone choice when transitioning from similarity-search to end-to-end detection paradigms.

1 Introduction

Historical documents constitute an invaluable source of information about our past as a society, offering insight into culture, identity, and history [1, 2]. However, the information conveyed by these documents is often difficult to analyze due to the degraded condition of the materials, the variability of handwriting styles, the use of archaic or extinct languages, and, no less importantly, the richness of the illustrations embedded throughout the documents, which not only provide an aesthetic finish consistent with the period in which they were produced but also convey vital information about the historical, social, and cultural context surrounding the text. A remarkable example of this phenomenon is the Codex Gigas, the largest surviving medieval manuscript, whose nearly full-page illustration of the Devil has fueled centuries of scholarly debate about the symbolic and historical meaning embedded in its pages [?].

Sketching is one of the oldest forms of communication developed by humankind, with evidence suggesting that humans have used drawings to depict their visual world since prehistoric times [4]. Unlike written language, a sketch does not require a shared vocabulary, alphabet, or grammar to be understood, since it conveys meaning through universally recognizable visual cues rather than symbolic encoding, which makes it an inherently cross-cultural and language-independent medium of expression [4, 5]. This universality, together with the widespread availability of touch-screen devices that make sketching as effortless as drawing with a finger, has motivated a growing body of research

on sketch-based image retrieval (SBIR), a paradigm in which a hand-drawn sketch, rather than an example image or a textual description, is used to query a database of images [6, 7, 8].

Currently, most historical documents are being digitized, and there are many efforts to analyze them using computer vision techniques; however, most of these techniques are focused on text recognition rather than on the analysis of the visual patterns present in their illustrations [9, 10]. A smaller but growing line of research, known as pattern spotting, addresses precisely this gap by searching for occurrences of a graphical pattern, such as an illuminated initial, a coat of arms, or a decorative motif, within a collection of historical document images [11, 12, 13, 14, 15]. Nevertheless, all of these approaches share a common constraint: they require an example image of the pattern of interest as the query, an assumption that does not always hold in practice, since researchers and archivists frequently know what they are looking for only as a mental image or a rough visual concept, without having an actual instance of it readily available.

Therefore, in this paper, we propose a sketch-conditioned, one-stage detector for pattern spotting in historical documents that builds upon [16] by reusing its domain-adapted encoder, iDoc, together with the more recent state-of-the-art encoder DINOv3 [17], as frozen backbones, while replacing the offline region-proposal and similarity-search decoder of our previous work with a trainable detection head conditioned directly on a hand-drawn sketch query through a combination of global feature-wise modulation and cross-attention with the document features. This design bridges the gap between pattern spotting and sketch-based image retrieval discussed above, allowing the system to localize a graphical pattern even when no example image of the target is available, and we further evaluate both encoders under this new conditioning scheme to determine which representation generalizes better to the graphical content of medieval manuscripts.

2 Related Work

2.1 Pattern Spotting in Historical Documents

Pattern spotting in historical documents was first formalized by En et al. [11, 12], who introduced both a BoVW-based retrieval system and the DocExplore benchmark used throughout this line of work. Building on deep learning, Úbeda et al. [13, 14] replaced hand-crafted descriptors with convolutional and feature-pyramid representations, improving robustness to scale variations, while Curi et al. [15] later achieved the strongest results to date through a dense correlation between query and document feature maps, at the cost of a high computational overhead. More recently, our previous work [16] addressed this limitation by combining a self-supervised, domain-adapted encoder (iDoc) with an open-set region proposal module, reaching state-of-the-art accuracy at a fraction of the search time; however, like all the methods above, it still relies on an example image of the target pattern as the query.

2.2 Sketch-Based Image Retrieval

Sketch-based image retrieval has evolved from hand-crafted edge-orientation descriptors such as HELO and S-HELO [7, 18] and mid-level keyshape representations [8], evaluated on the benchmark introduced by Eitz et al. [6, 4], to deep learning approaches such as Sketch-a-Net [19] and large-scale sketch-photo datasets like Sketchy [20], which enabled end-to-end cross-domain embeddings (see [5] for a comprehensive survey). Despite this progress, the application of sketches to historical or heritage document collections remains scarcely explored: to the best of our knowledge, the only prior attempt is the sketch-GAN of Creswell and Bharath [21], restricted to a single class of symbols (merchant marks) and a shallow generative architecture, leaving open the question of how modern self-supervised encoders and region proposal mechanisms can support sketch-based pattern spotting at scale.

2.3 Conditional and One-Shot Object Detection

Beyond retrieval-based pattern spotting, our detector is also related to one-shot object detection, where a single query exemplar is used to localize instances of an arbitrary, possibly unseen, class. Early methods cast this as a two-stage problem, augmenting Faster R-CNN-style proposals with non-local co-attention and channel-wise co-excitation between query and target features [22], or with hierarchical relation modules that compare query and candidate regions at multiple levels of granularity [23]. More recent work favors single-stage, anchor-free formulations: OS2D [24] performs dense correlation matching with a learned geometric transformation on top of an SSD-style anchor grid, while Li et al. [25] couple a Siamese FCOS backbone with a relation module to detect novel classes without any fine-tuning on the support image. Our detection head follows this anchor-free, FCOS-based line of work, but departs from prior conditioning strategies by combining a coarse,

global FiLM modulation with a fine-grained bidirectional co-attention, and by targeting graphical patterns expressed as a hand-drawn sketch rather than photographed as an exemplar crop.

3 The Proposal

Fig. 1 shows the general architecture of the proposed system, a one-shot detector that locates instances of a graphical pattern in a historical document page using a single hand-drawn sketch as the query, instead of relying on an example image of the target. The model couples a frozen, domain-adapted visual encoder (iDoc or DINOv3) with a Feature Pyramid Network and an FCOS detection head, which is conditioned on the sketch through two complementary mechanisms, a global FiLM modulation and a bidirectional co-attention between the sketch and the document features, allowing the network to both leverage coarse semantic cues and refine the search using fine-grained, query-specific spatial information.

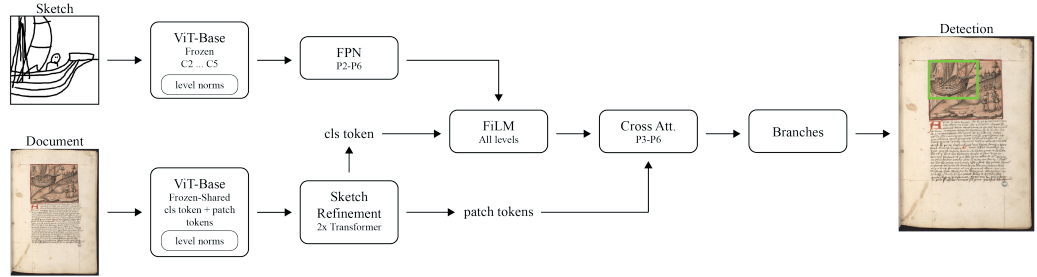


Figure 1: General architecture of the proposed sketch-conditioned pattern spotting model. A frozen ViT backbone (iDoc or DINOv3) encodes the document page into multi-scale features through an FPN, while the same backbone encodes the sketch query into a global embedding and a set of local patch tokens, refined by a lightweight transformer encoder. Both sketch representations condition the FCOS detection head through FiLM and bidirectional co-attention, producing class, box, and centerness predictions at every pyramid level.

3.1 Architecture

The backbone is a frozen Vision Transformer, either the iDoc encoder introduced in our previous work [16] or the more recent DINOv3 [17], both supported transparently by detecting the structure of the checkpoint at loading time. Multi-scale features are obtained by extracting the token sequences at four intermediate blocks of the transformer and reshaping them into spatial maps C_2 through C_5 , which are subsequently normalized by a small set of trainable LayerNorm layers to adapt the frozen representation to the detection task without modifying the backbone weights, and fed into a standard top-down Feature Pyramid Network [26] that produces the pyramid levels P_2 through P_6 used by the detection head.

The same frozen backbone is reused to encode the sketch query, producing a global class token and a set of patch tokens that are spatially pooled to a fixed grid to keep the cost of the subsequent attention operations manageable. Because the backbone was optimized for representing historical document patterns in general rather than for comparing a sketch against a specific image region, a lightweight, near-identity-initialized transformer encoder refines both the global and the local sketch representations before they are passed to the detection head, learning to emphasize the tokens that best discriminate the target shape from the background of the sketch.

The refined sketch representations condition the detection head through two complementary mechanisms. The global embedding modulates every pyramid level through a Feature-wise Linear Modulation (FiLM) layer [27], providing a coarse, class-level prior at a negligible computational cost, while the local patch tokens interact with the document features through a bidirectional co-attention module applied from an intermediate pyramid level onward: the sketch representation first attends to the document features to adapt itself to the visual style of the current page, and the document features then attend to this contextualized sketch, sharpening the spatial regions that resemble the query.

On top of this conditioned representation, a standard FCOS head [28] predicts, at every location of every pyramid level, a classification score, a bounding box expressed as the distance to the four sides of the object, and a centerness score that down-weights predictions far from the object center; the final detection score is the product of the sigmoid of the classification and centerness predictions.

The model is trained end to end with a combination of focal loss for classification, generalized IoU loss for box regression, and binary cross-entropy for centerness, complemented by a supervised contrastive loss [29] over the global sketch embeddings of the batch, which compacts the representation of sketches belonging to the same class and is particularly useful for the long tail of categories with very few annotated examples.

Training proceeds in two stages: the new sketch-conditioning modules are first optimized in isolation while the rest of the network remains frozen, and the whole detector is then fine-tuned jointly with a lower learning rate for the previously converged components, which we found to stabilize the introduction of the conditioning mechanisms without disrupting the pattern-spotting capability inherited from [16]. At inference time, an adaptive score threshold raises the operating point in densely decorated pages to suppress the long tail of weak activations, and a context-aware non-maximum suppression step further removes clusters of similarly-scored detections that are characteristic of repetitive textures rather than genuine objects; when more than one sketch example of the same class is available, their embeddings can also be averaged into a single prototype to reduce the influence of idiosyncratic strokes.

3.2 Dataset

We evaluate our proposal on a dataset of 544 historical document pages [COMPLETAR: fuente del corpus], annotated with 1,349 bounding boxes spanning 22 classes of recurring graphical patterns, with an average of 2.48 annotated instances per page and no invalid (zero-area) boxes found during the audit. Table 1 summarizes the class distribution. The class distribution is severely long-tailed (Gini index of 0.537): the most frequent class, *marqueur*, alone accounts for 32.2% of all instances (434 occurrences), while 17 of the 22 classes (77%) contain fewer than 50 labeled examples, with the two rarest classes counting only 7 instances each; this imbalance directly motivates the supervised contrastive loss and the class-aware copy-paste augmentation described above.

Table 1: Class distribution and median instance area in the annotated dataset (544 pages, 1,349 bounding boxes, 22 classes).

Class	N	%	Median area (px ²)
marqueur	434	32.17	741.0
S	138	10.23	1,272.5
losange	117	8.67	3,306.0
double_sep	98	7.26	1,572.5
croix	91	6.75	1,248.0
petit_A	49	3.63	1,512.0
encadrement	45	3.34	6,750.0
obj_1	39	2.89	11,016.0
pdp	35	2.59	1,200.0
status	34	2.52	16,371.0
obj_61	34	2.52	9,743.5
T	32	2.37	1,500.0
grand_A	31	2.30	3,700.0
obj_2	31	2.30	358,515.0
BP	28	2.08	19,773.0
obj_40	25	1.85	10,925.0
obj_34	24	1.78	31,010.5
triple_sep	20	1.48	3,960.0
obj_37	16	1.19	10,536.0
obj_39	14	1.04	7,198.0
obj_42	7	0.52	73,502.0
obj_31	7	0.52	30,504.0

Beyond class frequency, the annotated patterns also exhibit substantial geometric variability, as illustrated in Fig. 2: instance width and height range from 13 to over 3,000 px, with area spanning more than four orders of magnitude (from 234 px² to 3.1×10^6 px²); the median instance is small and moderately horizontal (50×28 px, aspect ratio 1.73). Following the standard COCO size convention, 30.7% of the instances are small (area below 32² px), 54.6% are medium (32² ≤ area < 96² px), and 14.7% are large (area ≥ 96² px); aspect ratios are skewed toward horizontal shapes, with 72.9% of instances above AR = 1.2 (35.0% of them extremely so, AR > 2), against 16.5% near-square instances and only 10.6% vertical ones.

This variability is intrinsic to the semantics of the illustrations themselves, since the size and aspect ratio of a graphical pattern depend on the type of object it represents, its relative importance

within the composition, and the amount of detail the scribe or illuminator devoted to it.

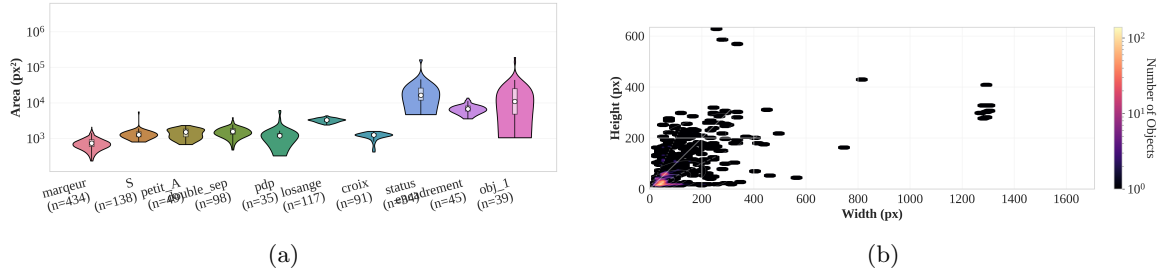


Figure 2: Geometric variability of annotated instances in the dataset. (a) Distribution of bounding box areas across all classes. (b) Joint distribution of bounding box width and height.

Beyond this geometric variability, the sketch-conditioned setting introduces three additional sources of difficulty that directly motivated the design choices described above: most classes have very few annotated instances, which limits the supervision available for learning a discriminative sketch representation; the same symbol can vary considerably in appearance across pages and scribes, so a single sketch may not be representative of every occurrence; and densely decorated pages contain repetitive textures and ornaments that are easily confused with the target pattern, leading to a large number of false positives if left unaddressed.

4 Experimental Results

We evaluate both backbone variants on the validation split (55 pages, 111 ground-truth boxes) at $\text{IoU} = 0.50$ and a score threshold of 0.25. Fig. 3 summarizes the comparison visually, and Table 2 reports the full set of global metrics.

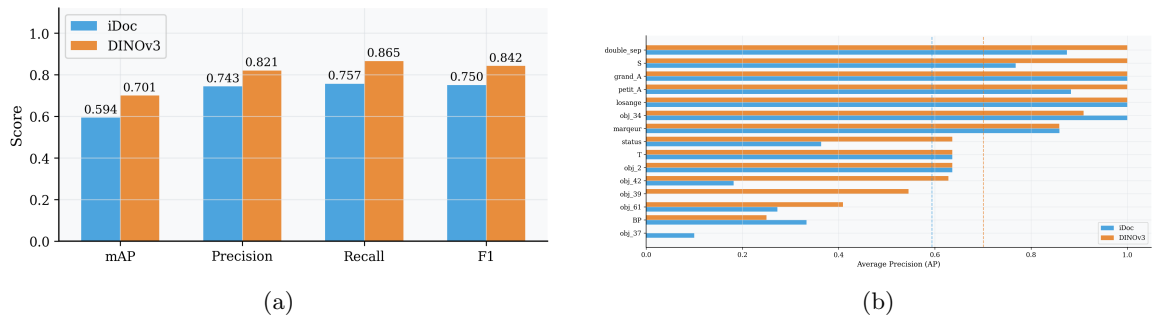


Figure 3: Comparison of iDoc-FCOS and DINOv3-FCOS on the validation split. (a) Global metrics (mAP, Precision, Recall, F1): DINOv3 outperforms iDoc across all four measures. (b) Per-class Average Precision (AP) sorted by DINOv3 performance; dashed vertical lines indicate the mAP of each backbone. DINOv3 dominates in rare and visually complex classes, while iDoc remains competitive or superior on a small number of classes (*obj_34*, *BP*, *obj_37*).

4.1 Global Metrics

Table 2: Global detection metrics for both backbone variants on the validation split (55 pages, 111 ground-truth boxes, $\text{IoU} = 0.50$, score thr. = 0.25). Δ denotes the absolute gain of DINOv3-FCOS over iDoc-FCOS.

Metric	iDoc-FCOS	DINOv3-FCOS	Δ
mAP @ $\text{IoU}=0.50$	0.594	0.701	+0.107
Precision	0.743	0.821	+0.077
Recall	0.757	0.865	+0.108
F1	0.750	0.842	+0.092
Total TP	84	96	+12
Total FP	29	21	−8
Total FN	27	15	−12

DINOv3-FCOS outperforms iDoc-FCOS across every reported metric. The mAP gap of +10.7 points is the most consequential difference: DINOv3 correctly localizes 12 additional instances (96 vs. 84 true positives) while simultaneously generating 8 fewer false positives (21 vs. 29), yielding an 8-point improvement in precision and a 10.8-point improvement in recall. This result is noteworthy because iDoc was explicitly adapted to the historical document domain via LoRA fine-tuning on the HORAE corpus [16], whereas DINOv3 is a general-purpose encoder with no exposure to medieval manuscripts during pre-training. The outcome suggests that, when the backbone is used as a *frozen* feature extractor for a trainable detection head, the representational breadth acquired during large-scale general pre-training transfers more effectively to the sketch-conditioned FCOS head than the more narrowly specialized representations of iDoc.

4.2 Per-Class Analysis

Table 3 reports the per-class AP for all 15 classes that have at least one ground-truth instance in the validation split; the remaining 7 classes are absent from this partition due to the dataset’s severe class imbalance (Gini index 0.537) and cannot be evaluated. Fig. 3(b) provides a visual overview sorted by DINOv3 performance, with dashed vertical lines marking the mAP of each backbone.

Table 3: Per-class Average Precision (AP) at IoU = 0.50 for the 15 classes with at least one ground-truth instance in the validation split. GT denotes the number of annotated boxes. The winner column indicates which backbone achieves higher AP; ties are marked with —. Best result per class in **bold**.

Class	GT	iDoc-FCOS	DINOv3-FCOS	Winner
grand_A	6	1.000	1.000	—
losange	4	1.000	1.000	—
T	3	0.636	0.636	—
obj_2	3	0.636	0.636	—
double_sep	8	0.875	1.000	DINOv3 (+0.125)
S	6	0.768	1.000	DINOv3 (+0.232)
petit_A	4	0.883	1.000	DINOv3 (+0.117)
status	4	0.364	0.636	DINOv3 (+0.273)
marqueur	60	0.859	0.859	DINOv3 (−0.001)
obj_61	4	0.273	0.409	DINOv3 (+0.136)
obj_42	2	0.182	0.628	DINOv3 (+0.446)
obj_39	2	0.000	0.545	DINOv3 (+0.545)
obj_34	3	1.000	0.909	iDoc (−0.091)
BP	1	0.333	0.250	iDoc (−0.083)
obj_37	1	0.100	0.000	iDoc (−0.100)
Classes with AP ≥ 0.50		8 / 22	12 / 22	

The per-class breakdown reveals a consistent pattern. DINOv3-FCOS wins in 7 of 15 evaluated classes, ties in 4, and loses in only 3. Its advantage is largest on rare and visually complex classes: *obj_39* (+0.545), *obj_42* (+0.446), *status* (+0.273), and *S* (+0.232), none of which exceed 34 annotated instances in the full dataset (see Table 1). This suggests that the broader feature vocabulary of the general-purpose backbone generalizes better precisely where training supervision is scarce, complementing the effect of the supervised contrastive loss that was specifically designed to address this long-tail regime.

Conversely, iDoc-FCOS retains a small edge on three classes (*obj_34*, *BP*, *obj_37*), all of which have very few instances in the validation split (1–3 ground-truth boxes), making these results statistically fragile. On the dominant class *marqueur* (60 ground-truth boxes in the validation split), both backbones are essentially equivalent ($\Delta = -0.001$), confirming that domain adaptation is beneficial when sufficient training signal is available, but does not provide a decisive advantage at this scale.

Finally, seven classes are absent from the validation split (*croix*, *encadrement*, *obj_1*, *obj_31*, *obj_40*, *pdp*, *triple_sep*), a direct consequence of the imbalanced class distribution. This motivates the adoption of stratified cross-validation in future work to ensure a reliable per-class estimate across all 22 categories.

5 Conclusions

We have presented a sketch-conditioned, one-stage detector for pattern spotting in historical documents that bridges the paradigms of sketch-based image retrieval and graphical pattern localization. By conditioning a standard FCOS detection head on a hand-drawn sketch through a combination of

FiLM modulation and bidirectional co-attention, the proposed system can localize recurring graphical patterns in medieval manuscript pages without requiring any example image of the target, relying solely on a query drawn by the user at query time.

The comparative evaluation of two frozen ViT backbones on a dataset of 544 annotated pages yields a clear and instructive finding: DINOv3-FCOS achieves a mAP of 0.701, outperforming iDoc-FCOS (0.594) by +10.7 points despite the fact that iDoc was explicitly adapted to the historical document domain. DINOv3 wins in 7 of 15 evaluated classes and ties in 4, while iDoc prevails in only 3, all of them with 1–3 ground-truth boxes in the validation split, making those victories statistically fragile. The advantage of DINOv3 is sharpest on rare and visually complex categories such as *obj_39* (+0.545), *obj_42* (+0.446), *status* (+0.273), and *S* (+0.232); on the dominant class *marqueur* (60 instances in the validation split), both backbones are virtually identical ($\Delta = -0.001$). This pattern suggests that domain adaptation is beneficial when ample supervision is available, but that the broader feature vocabulary acquired during large-scale general pre-training provides a decisive advantage in the low-resource regime that characterizes most classes in historical document collections. The implication for practitioners is that, when a backbone is used as a frozen feature extractor for a trainable detection head rather than as a direct similarity-search encoder, pre-training data diversity matters more than domain specificity.

Several directions remain open for future work. First, the severe class imbalance in the dataset (Gini index 0.537) leaves seven classes entirely absent from the current validation split; stratified cross-validation would provide a more complete and reliable per-class performance estimate across all 22 categories. Second, the contributions of the supervised contrastive loss and the copy-paste augmentation both motivated by the long-tail distribution warrant a dedicated ablation study to quantify their individual impact, especially for the classes where DINOv3 shows the largest gains. Third, the detection head currently conditions on a single sketch per query; extending the prototype averaging strategy to multiple sketches per class and studying how sketch diversity affects localization accuracy would make the system more robust to idiosyncratic drawing styles. Finally, lightweight end-to-end adaptation of the backbone within the detection pipeline for instance, through LoRA applied to the frozen ViT during joint training may further close the performance gap between the two encoder variants on rare classes while preserving the generalization advantages of large-scale pre-training.

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